

Tutorial 2: The Shape of Uncertainty & Your First Trial as a Data Scientist (STUDENT VERSION)

Theme: Data Types, Central Tendency, and Measuring Spread

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Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

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1 Mission Brief: Entering the World of Uncertainty

Why This Tutorial Exists

In Tutorial 1, you audited deterministic financial models (interest) where one input strictly mapped to one output. Now, you enter the real world of an actuary and data scientist: a world of **uncertainty, variance, and messy data**. You cannot use a simple equation to predict a single stock price or insurance claim. Instead, you must model the *shape* of the data.

Your role: You are no longer just calculating averages; you are an **Algorithmic Data Janitor** and a **Risk Auditor**.

Your product: A serious Exploratory Data Analysis (EDA) dashboard built via code in Google Sheets.

Your weapon: Detectors (data type classification, outlier sensitivity, center-vs-spread checks, and invariance signatures).

The Audit Mindset & Data Pipeline (Non-negotiable Today)

Every number you report must be defensible using the following pipeline and detectors:

Step 1: Ask a statistical question (demands variability, not a single fact).

Step 2: Identify data types (categorical vs numerical; discrete vs continuous).

Step 3: Clean the data (missing values, wrong units, impossible values).

Step 4: Summarise the distribution (center, spread, shape, outliers).

Step 5: Decide what is defensible (choose the right summary for the context).

Step 6: Communicate a claim with evidence (one claim, supported by metrics and visuals).

Mandatory Detectors:

- **Type detector:** Is this mathematical operation allowed for the data type?
- **Outlier/Robustness detector:** Does one extreme value completely distort the conclusion?
- **Center-vs-spread detector:** Same mean does not imply the same behaviour. Always check spread.
- **Invariance detector:** If you add a constant to every value, what changes? (Spread shouldn't).
- **Consistency detector:** Two mathematically equivalent methods must match.

2 Cognitive Trigger: The “Average” Illusion

Trigger Question 1: Two Classes, Same Mean

★ Core

Two classes took the same quiz (out of 100). Their **mean score is 70 in both classes**.

Non-routine challenge: Without computing any standard deviation, decide which statements can be simultaneously true:

- “Both classes performed the same.”
- “Both classes are equally consistent.”
- “Both classes need the same teaching intervention plan.”

Your answer must include: (1) one sentence distinguishing **typical value** vs **consistency**, and (2) one sentence naming a **spread measure** that can

separate the two classes.

Trigger Question 2: The Billionaire in the Room

★ Core

Imagine a small coffee shop with 9 university students. Their average (mean) net worth is \$2,000. Suddenly, Elon Musk walks in and sits down.

Non-routine challenge: Without calculating exact numbers, decide what happens to the mathematical average of the room, and what happens to the “typical” wealth of the room.

Your answer must include: (1) one sentence about the **mean** (and how it reacts to extreme values), and (2) one sentence about the **median** (and what it represents).

Pause-and-Think: The Weather vs. Climate Paradox

★★ Insight

City A and City B both have an identical **mean average temperature** of 20°C over the year. However, City A is a coastal paradise, and City B is a desert that freezes at night and boils at noon.

Write one line explaining how a mathematician can mathematically prove these two climates are completely different, despite identical means.

3 Structural Core: Essential Concepts and Mechanisms

Vocabulary With Meaning (Data Types)

Before calculating anything, you must know what your data is. Formulas crash if data types are wrong.

- **Categorical Data:** labels or names (e.g., blood type, car brand). *Mean is not allowed.*
- **Numerical Data:** numbers representing quantities.
- **Discrete Numerical:** countable integers (e.g., number of accidents, siblings).
- **Continuous Numerical:** measurable on an infinite scale (e.g., time, height, financial payout).
- **Distribution:** the pattern of data values (center, spread, shape).
- **Outlier:** a value far from most of the data, potentially heavily skewing the mean.

Central Tendency (The Center of Gravity)

- **Mean (μ or $\bar{x}$):** mathematical balancing point; $\frac{\sum x}{n}$; vulnerable to outliers.

- **Median:** the 50th percentile rank; resistant to outliers.
- **Mode:** most frequent observation; crucial for categorical data.

Dispersion Mechanisms (Measuring Risk and Volatility)

- **Range:** max minus min; fast but uses extremes only.
- **Variance (σ^2):** average of *squared* differences from the mean.
- **Standard Deviation (σ):** square root of variance; returns to original units.
- **Interquartile Range (IQR):** $Q_3 - Q_1$; spread of middle 50%, resistant to extreme outliers.

One Sentence That Governs This Tutorial

Mean tells you where the center is; dispersion tells you how dangerous it is to trust that center.

4 Proof Decision Points: Navigating Logical Turns

Decision A: Which “Typical” Value Should You Use?

★ Core

A dataset is **right-skewed** because a few students have extremely high screen time.

Choose exactly one statement that is most defensible, then justify it in 3–5 sentences:

- (1) Mean is always the best typical value.
- (2) Median is often more representative when there are outliers.
- (3) Mode is always the best typical value for numerical data.

Decision B: The Algorithmic Breakpoint (Type Governance) ★★ Insight

A student claims: “The average phone brand in our class is iPhone.” At the same time, you write a script to average 5 insurance payouts: [100, 200, "100", 50, 150]. Notice the third value is a string.

Explain why both of these represent a **data type mistake**. What operation is allowed, and what happens if a computer tries to mathematically add a string?

Decision C: Why Do We SQUARE the Differences?**★★ Insight**

Variance uses $\frac{\sum (x_i - \bar{x})^2}{N}$. Why square $(x_i - \bar{x})$ rather than add raw differences?

Your answer must explain what happens if you don't square them.

Decision D: Range vs IQR vs Standard Deviation

★★ Insight

You are comparing two groups with the same median. You want a spread measure that is **not dominated by one extreme outlier**.

Classify each measure as: highly sensitive / moderately sensitive / resistant.

- Range
- IQR
- Standard deviation

Then write one sentence: “Which one would you use if you expect outliers, and why?”

5 Deepening Layer A: Calculation Micro-Labs

5.1 Micro-Lab 1: Data Types and Statistical Questions

1.1: Classify Variables (Fast and Correct)

★ Core

For each variable, classify it as: **categorical** or **numerical (discrete/continuous)**.

V1: Number of messages received yesterday

V2: Daily screen time (minutes)

V3: Favourite game genre

V4: Number of siblings

V5: Height (cm)

V6: Commute method (MTR / bus / walk / other)

1.2: Is It a Statistical Question?

★★ Insight

A statistical question anticipates variability and can be answered by data. Classify each as: statistical / non-statistical. Rewrite non-statistical ones to be statistical.

Q1: What is the exact height of our teacher?

Q2: How many minutes of screen time do students in our class have on weekdays?

Q3: Which phone brand is the best?

Q4: Do students in our class tend to sleep less than 7 hours on school nights?

5.2 Micro-Lab 2: Center, Spread, and Constraints

2.1: Reverse Engineering the Mean (Control the Constraints) ★ Core

You know the **answers** first. You must find the missing data. A student took 5 tests. The **mode** was 85. The **median** was 85. The **mean** was exactly 80. The tests are scored out of 100 as integers.

Find a set of 5 scores that makes this mathematically true. Show your reasoning.

2.2: Compute Mean, Median, Mode

★ Core

A group recorded weekday screen time (minutes) for 9 students:

25, 35, 40, 40, 50, 60, 60, 65, 85

- (a) Compute the mean.
- (b) Compute the median.

(c) Find the mode(s).

(d) Give one sentence: which is the best “typical” value here, and why?

2.3: Hand-Calculating Volatility (Variance)

★★ Insight

Two stocks over 4 days show the following price changes (%): **Stock X**: [2, 4, 6, 8]

Stock Y: [4, 5, 5, 6]

Q1: Without calculating, which stock has higher volatility (dispersion)? Why?

Q2: Verify mathematically. Calculate the variance σ^2 for Stock X.

(Step 1: mean; Step 2: squared deviations; Step 3: average.)

2.4: Range and IQR

★★ Insight

Using the screen time dataset:

25, 35, 40, 40, 50, 60, 60, 65, 85

- (a) Compute the range.
 - (b) Compute Q_1 , Q_3 , and IQR using the “median-of-halves” method (exclude the overall median).
 - (c) Use the $1.5 \times \text{IQR}$ rule to check for outliers.
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2.5: The Outlier Stress Test (Mean vs Median vs IQR)

★★ Insight

Replace the last value 85 by 150 (a data entry mistake):

25, 35, 40, 40, 50, 60, 60, 65, 150

- (a) Recompute mean and median. Which one changed more?
- (b) Which detector catches the mistake fastest: type / range / IQR / sanity check?
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6 Deepening Layer B: Building Concepts by Auditing Data

6.1 The Data Janitor: Cleaning Messy Inputs

Investigation 1: The Insurance Claim Database

★ Core

You receive a spreadsheet of claims. Column A is “Age”, Column B is “Claim Amount (\$)”.

Row	Age	Claim Amount (\$)
1	45	1200.50
2	“Thirty”	500
3	-5	3000
4	22	“N/A”
5	60	1,500.00

Task: Identify three distinct errors that break “continuous numerical data” or basic logic, and explain how a script must fix them before computing standard

deviation.

Investigation 2: Case File: “Screen Time and Focus”

★ Core

You receive a dataset from a student survey. It contains messy entries that will break any algorithm.

Variables: S : screen time (mins), F : focus rating (1–10), D : device (phone/tablet/laptop)

Raw Data:

ID	S (minutes)	F (1–10)	D
1	45	7	phone
2	60	6	phone
3	30	8	tablet
4	90	5	laptop
5	120	4	phone
6	2h	6	phone
7	75	11	tablet
8	–	7	phone
9	40	8	laptop
10	55	6	phone

Task: Identify at least 3 issues and label each as: missing / wrong unit /

impossible value / type error.

Investigation 3: The Cleaning Ledger (Make Choices Auditable)

★★

Insight

Create a cleaning ledger with three columns: **Problem**, **Action**, **Justification**. Detail exactly how you would process the errors found above before running statistics. Include at least one action for each: wrong type, impossible value, missing value, formatting.

Investigation 4: First Defensible Summary

★★ Insight

After cleaning the screen time data, compute for S :

- mean, median
- range, IQR

Then write a 5–7 sentence mini-report deciding which center measure is more defensible.

7 Extension Layer A: Real-World Case Study - Hong Kong Museum Attendance

The Museum EDA Pipeline

You are given a live CSV dataset of Hong Kong Museum Attendance containing thousands of records. The data is notoriously messy: numbers have commas, missing data is represented by hyphens, and remarks contain commas that break naive algorithms. Your task is to use Google Gemini to write Google Apps Scripts (GAS) that automatically clean, restructure, and analyze this data.

7.1 Phase 1: The Algorithmic Data Janitor

Google Gemini Prompt: Fetch and Clean the Raw Data

★ Core

For a deeper dive into coding a Data Janitor, try this with Google Gemini: "Act as an expert Google Apps Script developer. I need a script to read a messy CSV file from a URL and output a neat, tidy dataset into my Google Sheet.

Requirements: 1) Fetch the CSV directly from a URL. 2) Parse the CSV robustly, accounting for commas inside quotation marks (e.g., in the remarks).

3) Clean the 'Attendance' column: trim whitespaces, remove thousands-separator commas, and replace hyphens '-' (representing closed museums) with empty strings. 4) Write the cleaned data into a new sheet named 'Cleaned Museum Data'."

Audit 1: Verify the Cleaning Ledger

★ Core

After running the generated script, inspect the "Cleaned Museum Data" tab.

- **Type Check:** Click on an attendance number (e.g., 280488). Is it treated as a number or a string?
- **Missing Values:** How did the script handle Q4 2018 for HKMCD?

7.2 Phase 2: Data Restructuring (Pivoting)

Google Gemini Prompt: Pivot the Dataset for Time-Series Analysis ** Insight

To explore structural transformations, use this prompt with Gemini: "Now that I have a 'Cleaned Museum Data' sheet, write a new Google Apps Script to pivot this data.

Requirements: 1) Read the cleaned data array. 2) Create a cross-tabular Pivot Sheet named 'Attendance by Museum'. 3) Combine "Year" and "Quarter" into a single row heading (e.g., "2018 Q1"). 4) Make the unique Museum Venue Codes the column headings. 5) Map the attendance numbers to the correct intersections, leaving cells blank if no data exists. 6) Freeze the header row and the first column for easy scrolling."

7.3 Phase 3: Statistical Dashboards & Dispersion Charts

Google Gemini Prompt: Automated Statistical Analysis & Box Plots *** Extension

To automate statistical summaries, paste this prompt into Gemini: "Write a final Google Apps Script that acts as an automated data analyst for my pivoted

museum data.

Requirements: 1) Loop through every unique museum in the dataset. 2) For each museum, create a new sheet (e.g., 'Stats - HKMA'). 3) Instead of just computing the numbers, inject actual Google Sheets formulas (using setFormula) into the cells to calculate: Mean (=AVERAGE), Median (=MEDIAN), Min (=MIN), Max (=MAX), Sample Std Dev (=STDEV.S), and Quartiles (=QUARTILE). 4) Use the 5-number summary (Min, Q1, Median, Q3, Max) to automatically insert a Candlestick Chart (which acts as a Box and Whisker diagram) into each sheet."

Audit 2: Spread and Risk Detectors

*** Extension

Go to the generated **Stats - HKMA** sheet and examine the Candlestick chart alongside the formulas.

A1: Center vs Spread: Compare the Mean and the Median. Is the data skewed? How does the Candlestick chart visually prove this skew?

A2: Invariance Test: Manually add 10,000 to every attendance record in column B. Watch the injected formulas recalculate. Does the **STDEV.S** change? Does the **IQR** change? Explain why spread measures are invariant to addition.

8 Extension Layer B: Pressure-Testing Data Claims Like an Actuary

8.1 Claim Battles: Choose the Right Summary

Claim Battle 1: “The Typical Screen Time Is 70 Minutes” **** Insight**

A student claims: “The typical weekday screen time is 70 minutes.” They used the mean.

Task: Write a structured response:

- One sentence: whether mean is defensible here, and under what condition.
- One sentence: provide an alternative (median) and justify it against skew.

Claim Battle 2: “Our Class Is Consistent” **** Insight**

A student claims: “Our class is consistent because the range is only 90 minutes.”

Task: Explain why this reasoning can be weak. Your answer must mention: **(i) extremes, (ii) middle 50%, and (iii) context.**

8.2 Bridge to Risk: Why Spread Is a Risk Signal

Risk Bridge: Same Center, Different Risk

*** Extension

Two insurance products have the same expected payout (same mean), but one has a much larger standard deviation.

Explain in 3–4 sentences:

- Why the larger spread product is “riskier” even with the same mean.
 - What kind of customer might still prefer it.
 - Name one detector you would use before buying.
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The Skewed Distribution Trap (Venture Capital)

*** Extension

A venture capital firm invests in 100 tech startups. 99 of them go to \$0. One becomes the next Google, returning \$10 Billion.

Question: Why are standard “Mean” and “Standard Deviation” terrible metrics for explaining this dataset? Explain using asymmetry (skew). Then state the median in one sentence and explain why it is a more honest summary.

9 Conclusion: Finalizing the Audit Output

Your Final Output for Day 2

★ Core

By the end of today, you should have:

- (P1) An **EDA Trial Lab** spreadsheet with a Raw tab, Clean tab, and Dashboard tab.
- (P2) A working Google Apps Script calculating your custom standard deviation formulas.
- (P3) A dashboard showing a stock price moving inside its standard deviation “Bollinger Bands”.
- (P4) A written **cleaning ledger / audit summary** showing what you changed and why data cleaning is the most important step before applying statistical formulas.
- (P5) A one-paragraph **data claim** supported by at least two summaries (center + spread).
- (P6) Evidence of responsible AI use: at least **two** prompts + **four** detector checks.

Exit Ticket: The One-Sentence Rule

Complete this sentence (must include one of: data type / invariance / center vs spread):

“A data summary is trustworthy only if

emph”

10 Optional Challenges: Customizing Your Learning Depth

★ Core: Clean Calculation + Clean Interpretation

C1: Median over Mean: Name a real-world public policy statistic (besides wealth) where reporting the median is much more honest than reporting the mean.

C2: Construct a dataset where mean = median = mode, but the data is still very spread out.

C3: Construct a dataset where IQR suggests “stable” but range suggests “unstable”.

★★ Insight: Detect Hidden Assumptions

I1: The $n - 1$ Correction (Sample vs Population): If you audit a small sample of 10 students rather than the entire school, statisticians divide variance by $(n - 1)$

instead of n . Why does dividing by a smaller number make the final variance larger? Conceptually, why do we want a *larger* estimate of risk when looking at a small sample?

I2: Design a “red flag” checklist: give 4 signs a dashboard is wrong even if it prints numbers.

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